CASE STUDY | SUPPLY CHAIN AND LOGISTICS ENGINEERING | CHAMPIONSHIP

PortFlow Container Terminal: Yard and Gate Flow

This is a flow problem that looks like a space problem. Judges test whether you separate causes from symptoms with the case data (Little's Law on dwell), resist the obvious expansion, and still cover finance, operations and safety, supply chain, change management and AI within a $10M budget.

THE ANSWER

"Option B — Integrated flow package: shared visibility, appointments, AI-assisted yard planning — delivers 21.7% lower year-5 yard utilization (target 19.5%) for CAD $9,000,000 of the CAD $10,000,000 budget: NPV CAD $6,491,191 at 9.5%, payback 31.7 months. Five-year net financial impact (case formula): -CAD $16,500,000 (net savings)."

19.5%	$10.00M	12 mo	$6.49M
TARGET	**BUDGET CEILING**	**DEADLINE**	**NPV OF ANSWER**
Lower year-5 yard utilization	capital, CAD	core implementation	at 9.5% WACC

INSIDE

01 The Challenge
02 The Approach
03 The Options
03b The Package
04 The Results
05 Execution
05b Required Analysis
06 Defence & Submission

Archetype: Operations & engineering improvement · Family: Throughput / capacity · Model: Baseline KPIs → bottleneck/root cause → optimize/simulate → pilot → ROI + KPIs

SITUATION · COMPLICATION · QUESTION

What this case is really asking

This is a flow problem that looks like a space problem. Judges test whether you separate causes from symptoms with the case data (Little's Law on dwell), resist the obvious expansion, and still cover finance, operations and safety, supply chain, change management and AI within a $10M budget.

S SITUATION

PortFlow is a Canadian container terminal linking shipping lines, trucking, rail and inland customers. Today the yard holds 21,120 TEU against 24,000 TEU effective capacity (88%), containers dwell 6 days, and trucks take 72 minutes per visit (40 queuing, 32 inside). Volume will grow 20% in five years. The terminal spends $6M a year on overtime and $8M on delay and service penalties. Release and schedule information is not shared consistently, planning runs across separate systems on fixed rules, and the yard system knows a container's stack but not its exact position. One 15-acre parcel (about 4,000 TEU) could be added, phased around live operations.

C COMPLICATION

At +20% volume, the same 6-day dwell would need 106% of yard capacity, and trucks already take 72 minutes against a 45-minute objective. Overtime ($6M) and penalties ($8M) are rising, and the one expansion parcel would only bring utilization to 90.5%.

Q QUESTION

Which package, within the $10M upfront budget, attacks PortFlow's core issue so that at +20% volume the yard runs at or below 85% utilization and trucks turn in under 45 minutes by year five?

A ANSWER

Option B — Integrated flow package: shared visibility, appointments, AI-assisted yard planning — delivers 21.7% lower year-5 yard utilization (target 19.5%) for CAD $9,000,000 of the CAD $10,000,000 budget: NPV CAD $6,491,191 at 9.5%, payback 31.7 months. Five-year net financial impact (case formula): -CAD $16,500,000 (net savings).

CONSTRAINTS

- Upfront investment at most $10M CAD; it does not need to be spent i...
- Year-5 objectives: average yard utilization <= 85% and average truc...
- Vessel and truck operations cannot stop while changes are introduced
- Customs holds and missing documentation cannot be overridden by Por...
- Up to 8 content slides for a 10-minute presentation

WHAT THE JUDGES SCORE



SOLUTION MODEL · OPERATIONS & ENGINEERING IMPROVEMENT

How we cracked it

Baseline KPIs → bottleneck/root cause → optimize/simulate → pilot → ROI + KPIs

How does PortFlow reach <= 85% yard utilization and < 45-minute truck turns at +20% volume, within $10M?

Yard inventory (dwell)

- Why do containers stay 6 days: late release information, uncollected imports, early exports or empties?
- What dwell meets 85%? (4.83 days on 24,000 TEU)

Yard handling (rehandles)

- How many crane moves are reshuffles because exact positions and pickup times are unknown?
- Can stacking by pickup time free RTG hours for the 20% growth?

Gate and landside (arrivals)

- Why do trucks bunch: no appointments, no readiness signal?
- What cuts the 40-minute queue and the 32 minutes inside?

HYPOTHESIS (FORMED IN THE FIRST 20 MINUTES)

"The core issue is fragmented release and schedule information, not a lack of space. Sharing it, booking trucks against ready containers and planning the yard around known pickups cuts dwell below 4.83 days and truck turn below 45 minutes without new land."

TOOLKIT

- Little's Law: inventory = throughput x dwell
- Cause-versus-symptom tree (5 whys)
- Queueing view of truck arrivals: peak-to-average ratio
- Decision matrix with budget as a hard gate

QUICK MATH

- Today: 21,120 TEU / 6 days = 3,520 TEU/day; 21,120 / 24,000 = 88%
- Year 5: 4,224 TEU/day x 6 days = 25,344 TEU = 106% of capacity
- Expansion only: 25,344 / 28,000 = 90.5% (misses 85%)
- Dwell needed: 0.85 x 24,000 / 4,224 = 4.83 days (-19.5%)
- Truck turn: 40 + 32 = 72 min; target < 45 needs -27 min or more

TIME PLAN FOR A 3-HOUR CRACKING WINDOW

0:00-0:25	0:25-0:55	0:55-1:40	1:40-2:30	2:30-3:00
Read & align	Quick math	Package & finance	Deck (8 slides)	Rehearse & Q&A
SCQA, cause-vs-symptom tree, first hypothesis	Little's Law, dwell threshold, truck turn gap	Initiatives, costs, owners, 5-year formula	Answer first, matrix, budget check, plan, risks	10-minute run, likely judge questions

SOLUTION LADDER

Four levels of solution, one recommendation

Option B is recommended. It is both the cheapest compliant option and the best value: a dominant choice.

LEVEL 1 · QUICK WIN

Option A

10% · $4.10M
Lowest capital: what a minimal move buys

LEVEL 2 · MINIMUM COMPLIANT

Option B

21.7% · $9.00M
Cheapest option that meets the target within budget

RECOMMENDED

LEVEL 3 · VALUE MAXIMISER

Option B

21.7% · $9.00M
Highest NPV within budget

RECOMMENDED

LEVEL 4 · STRETCH

Option B

21.7% · $9.00M
Largest improvement regardless of budget

RECOMMENDED

DECISION MATRIX (SCORE 0-10, WEIGHTED)

Criterion	Weight	Option A	Option B (chosen)	Option C
Target attainment	30%	3.4	7.4	4.9
Financial return (NPV)	25%	9.1	10.0	2.0
Payback speed	15%	5.2	4.7	0.0
Downside resilience (-15% perf.)	15%	4.4	9.5	6.2
Budget headroom	15%	5.9	1.0	0.0
Weighted total	100%	5.6	7.0	2.9

THE THREE ALTERNATIVES

Option A: Gate quick win: truck appointments and release portal

$4.10M · 10% · NPV $4.16M
In budget · misses target

Option B: Integrated flow package: shared visibility, appointments, AI-assisted yard planning

$9.00M · 21.7% · NPV $6.49M
In budget · meets target

Option C: Yard expansion on the 15-acre parcel with two added RTGs

$14.50M · 14.3% · NPV -$14.63M
OVER BUDGET · misses target

RECOMMENDED PACKAGE

Integrated flow package: shared visibility, appointments, AI-assisted yard planning

Every initiative has an upfront cost, an ongoing cost and a named owner for its savings.

INITIATIVE & PURPOSE	UPFRONT	PER YEAR	OWNER
Truck appointment system with gate pre-advice and OCR Spread arrivals into booked slots, only for released and ready containers; faster gate transactions.	$2.20M	$300K	Gate and Landside Manager
Release and ETA visibility portal One shared status for lines, truckers, rail and customers, with customs-hold flags and a named escalation contact.	$1.40M	$250K	Commercial Director
Slot-level positioning and AI-assisted stacking RTG position sensors plus a planning upgrade so boxes due out first sit on top; planners approve each recommendation.	$2.90M	$400K	Yard Planning Manager
Integrated operations planning cell One shared plan across vessel, yard, gate and rail teams, so a schedule change is re-planned once, not in several systems.	$800K	$200K	Operations Manager
Dwell-time rules: free-time tariff and empties evacuation Storage charges after free days, early-pickup incentives and agreements with lines to remove empties.	$200K	$0	Commercial Director
Change management and paid training Co-design workshops, super-users per shift and training in paid working time.	$700K	$150K	HR Director with the Employee Representative
Contingency (about 10%) Released at the phase-3 gate if unused.	$800K	$0	Finance Director

BUDGET CHECK



$9.00M committed (90%) **$1.00M headroom · budget $10.00M**

FIVE-YEAR NET FINANCIAL IMPACT (CASE FORMULA)

$9.00M + 5 x ($1.30M - $6.40M) = -$16.50M

Investment + 5 x (additional annual costs - annual savings); undiscounted; negative = net savings. Discounted cross-check: NPV $6.49M at 9.5%, payback 31.7 mo.

DIRECTIONS CONSIDERED · SAME FORMULA

Option A: Gate quick win: truck appointments and release portal
Set aside: fixes the gate queue but not the yard; dwell falls only about 10%, so year-5 utilization stays near 95% and rehandles keep trucks inside too long.
-$9.40M
upfront $4.10M

Option B: Integrated flow package: shared visibility, appointments, AI-assisted yard planning
Recommended.
-$16.50M
upfront $9.00M

Option C: Yard expansion on the 15-acre parcel with two added RTGs
Set aside: costs about $14.5M (over budget) and still misses the objective at 90.5% utilization; it treats the symptom while dwell keeps growing. Kept as a phase-2 option.
$13.50M
upfront $14.50M · over budget

ECONOMICS · PROFIT-DRIVEN

What the recommendation is worth

Run-rate benefit $6.40M/yr net of $1.30M/yr O&M; levelized 128752.43 CAD per year per % point of improvement.

$6.49M	29.9%	31.7 mo	1.46x
NET PRESENT VALUE	IRR	PAYBACK	BENEFIT / COST
5 yrs @ 9.5%	annual	monthly cash model	present value

CASH FLOW BY YEAR (BARS) AND CUMULATIVE (LINE)



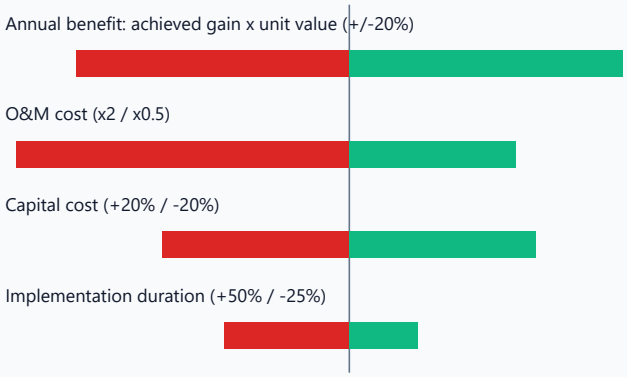
SCENARIOS

Bear (-15% perf., +20% capex)
18.45% · NPV $620K · payback 45.2 mo · misses target

Base (approved plan)
21.7% · NPV $6.49M · payback 31.7 mo · meets target

Bull (+15% perf., -10% capex)
24.95% · NPV $10.96M · payback 25.7 mo · meets target

NPV SENSITIVITY (TORNADO)



30 / 60 / 90 DAYS

From pilot to control

One owner per workstream; every phase ends in a go/no-go gate.

DAYS 1–30

Validate & pilot

- Confirm baseline data and the binding constraint
- Launch pilot of Option B at the constraint
- Name owners; start KPI dashboard

DAYS 31–60

Prove & decide

- Pilot gate: Slot adherence >= 80% on pilot lanes and no safety incidents
- Lock fixed-price packages for scale-up
- Train operators; write standard work

DAYS 61–90

Scale & control

- Begin scale appointments, positioning and ai advice
- Control chart on the primary KPI
- Review vs target (19.5%) with sponsor

IMPLEMENTATION HORIZONS

Phase 1 · Months 1-3 **Co-design and pilot**	KPI: Pilot lanes: queue time -25%; release status shared within 1 hour; baseline rehandles measured Gate: Slot adherence >= 80% on pilot lanes and no safety incidents	**$2.00M**
Phase 2 · Months 4-9 **Scale appointments, positioning and AI advice**	KPI: Dwell <= 5.3 days; truck turn <= 55 min; rehandles per box -20% Gate: AI advice accepted >= 70% with every override logged; tariff announced with 90 days notice	**$5.20M**
Phase 3 · Months 10-12 **Stabilise, connect rail and customers, hand over**	KPI: Dwell <= 4.8 days; truck turn < 45 min; utilization on track for <= 85% at year 5 Gate: Finance confirms savings against each owner's target; contingency released	**$1.80M**

TOP RISKS

Truckers do not adopt appointments (Medium / High)

Mitigation: Slot rules co-designed with trucking firms, grace windows, priority lanes for on-time trucks, a walk-in lane kept during transition
Trigger: Slot adherence < 75% for 2 weeks

Dwell does not fall enough (Medium / High)

Mitigation: Free-time tariff, release alerts, empties evacuation agreements; parcel design kept ready as phase 2
Trigger: Dwell > 5.3 days at month 9

Cutover disrupts live operations (Medium / High)

Mitigation: Lane-by-lane and block-by-block rollout, parallel running, go-lives outside vessel peaks, rollback plan
Trigger: Gate throughput below baseline for 2 shifts

KPIS: BASELINE → TARGET

Average yard dwell time

6.0 days → <= 4.8 days (plan 4.7)

Year-5 yard utilization at +20% volume

88% today; 106% if nothing changes → <= 85% (plan about 83%)

Truck turn time

72 min (40 queue + 32 inside) → < 45 min (plan about 40)

Rehandles per container

measured in phase 1 → -30%

Overtime and penalties

$6M and $8M a year → $3.6M and $4.0M a year

Yard and gate safety incidents

current rate → no increase; near-misses reviewed weekly

WHAT THE BRIEF ASKS FOR

Required analysis, area by area

Operations and safety, supply chain, change management, and AI and automation, each answered from the case data.

OPERATIONS & SAFETY

• Storage: dwell 6.0 to 4.7 days keeps year-5 inventory near 19,900 TEU (83% of 24,000) at +20% volume, without new land.
• Container movement: exact slot positions and stacking by planned pickup time cut rehandles, so RTG hours go to productive moves and the 12 RTGs absorb the 20% growth.
• Truck service: appointments spread arrivals (queue 40 to about 14 min); pre-advice and pre-staged containers shorten time inside (32 to about 26 min): about 40 minutes in total, below the 45-minute objective.
• Safety: fewer rehandles and shorter queues reduce truck-crane interaction; hard rules for hazardous, reefer and weight segregation stay locked; every go-live has a safety review.
• Measures, reviewed weekly: utilization, dwell, rehandles per box, queue and inside time, slot adherence, safety incidents and near-misses.

SUPPLY CHAIN

• Shipping lines: ETA changes feed one shared plan; export receiving windows and empty-container evacuation agreed.
• Truckers: book pickup slots only when the container is released and ready; live status and a guaranteed window.
• Rail: rail loads pre-staged by train departure; the rail operator sees release status through the same portal.
• Customers: release, hold and ETA alerts to plan replenishment, plus a named contact and escalation process for urgent shipments.
• Customs and documentation holds stay outside PortFlow's control, but they are flagged early so no truck is booked against a held container.

ORGANIZATIONAL CHANGE MANAGEMENT

• Stakeholders: yard operators, gate clerks, planners and supervisors, maintenance, trucking firms, rail, shipping lines, customers and finance.
• Involve staff before tools arrive: operators and planners co-design stacking rules and appointment rules in phase 1.
• Train in paid working time: super-users on every shift, simulator sessions, refresher training budgeted at $0.15M a year.
• Clarify roles: no layoffs from this program; gate clerks and planners move into the planning cell and exception handling; one RACI per process.
• Keep operations running: lane-by-lane and block-by-block rollout, parallel running, go-lives outside vessel peaks and a rollback plan.

AI & AUTOMATION

• Use: AI-assisted stacking and housekeeping recommendations: where to place each incoming container and which overnight moves to make, so boxes due out first sit on top.
• Information needed: exact slot positions (RTG sensors), appointment bookings, release and hold status, vessel and rail schedules, and container attributes (size, weight, reefer, hazardous).
• Oversight: the yard planner reviews and approves every recommendation; hard safety rules cannot be overridden by the model; the Yard Planning Manager owns model performance.
• Performance check: 6 weeks in shadow mode against the current rules, then weekly tracking of rehandles per box, acceptance rate and override reasons; retrain or roll back if rehandles do not fall.
• Set aside: fully automated cranes cost far more than the $10M budget, add safety and labour risk, and do not fix the information gap.

BEFORE YOU SUBMIT

Judge questions and the final checklist

14 prepared answers, all computed from the model. Answer first, then evidence.

STRESS TEST

Why not just build on the 15-acre parcel?

Space alone fails the objective: 25,344 TEU / 28,000 TEU = 90.5% at year 5. It costs about $14.5M, over the $10M budget, and must be phased around live operations. Cutting dwell to 4.7 days reaches about 83% on today's footprint. We keep the parcel as a phase-2 option if dwell stays above 5.6 days.

STRESS TEST

What if dwell does not fall as much as you assume?

The threshold is explicit: dwell at or below 4.83 days meets 85% without land. If dwell is above 5.3 days at month 9 (our gate trigger), we tighten the free-time tariff and start the parcel design. With dwell at 5.6 days or less, the parcel would then meet the objective.

STRESS TEST

How confident are you in the $6.4M of annual savings?

Each saving has an owner and a mechanism. Break-even needs only $3.1M a year. If penalties fall 25% instead of 50%, the five-year impact is still -$6.5M.

STRESS TEST

Will the AI and automation cost jobs?

No layoffs from this program. The AI recommends and planners decide. Roles shift from manual re-planning and overtime to planning-cell and exception work, with training in paid time.

STRESS TEST

How do you keep the terminal running during the rollout?

Lane-by-lane appointments with a walk-in lane kept open, block-by-block positioning, 6 weeks of shadow mode for the AI, go-lives outside vessel peaks, and a rollback plan triggered if gate throughput drops below baseline for 2 shifts.

STRESS TEST

How will you get truckers to use appointments?

We co-design slot rules with them, guarantee the container is ready when a slot is booked, give on-time trucks a priority lane, and keep grace windows. Slots exist only for released containers, so a booking is never wasted.

PRE-SUBMISSION CHECKLIST

✓ Answer-first executive summary (one sentence, three numbers)	Option B — Integrated flow package: shared visibility, appointments, AI-assisted yard planni…
✓ Three options costed on the same basis	decision matrix with weights shown
✓ Recommendation fits the budget	CAD $9,000,000 of CAD $10,000,000
✓ Recommendation meets the target	21.7% vs 19.5%
✓ Financials: NPV, payback, sensitivity	NPV CAD $6,491,191, payback 31.7 months
✓ Bear case tested (-15% performance, +20% cost)	bear NPV CAD $620,445
✓ 30/60/90 roadmap with owners and gates	pilot → prove → scale
✓ Top risks with triggers	5 risks logged
✓ Q&A bank of 10+ prepared answers	see judge_QA_preparation.md
! Assumptions validated with real data	replace: Monetisation, Discount rate (WACC), Appraisal horizon, Annual O&M, Benefit during i…
✓ Safety & licensed engineering review noted	MOC + HAZOP before commissioning

WHERE THIS CASE STYLE APPEARS (DIRECTORY-LISTED; WINNERS NOT VERIFIED)

Boeing Northwest Case Competition · USA
Colorado State University Supply Chain Case Competition · USA
Carlson National Undergraduate Supply Chain Case Competition · USA
University of St. Thomas Supply Chain Case Competition · USA
Kohler Undergraduate Wisconsin Supply Chain Cup · USA